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Dataset Acquisition



LVIS v1.0

1,203 categories · long-tail



COCO-ZSD

Generalized zero-shot recognition evaluation



Long-tail distribution



Head Common Rare

Class frequency distribution

2

Data Preprocessing



Image Resize & Normalise

ImageNet stats normalisation



Annotation Parsing

Annotated object regions · class labels



Evaluation protocol setup



Frequency-Aware Semantic Weighting

Region-Based Semantic Fusion Framework

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Semantic Prompt Generation



WordNet Synonyms

Hyponyms · hypernyms hierarchical relations



LLM Paraphrasing

GPT-based diverse descriptions

cat

mouser

tabby

Kitty

feline

domestic cat

1–10 prompts per category

Complementary semantic prompt sources

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Frequency-aware semantic weighting



Image Stream

CLIP-ViT encoder

Visual feature embeddings

Annotated object regions



Text Stream

CLIP text encoder

Per-prompt text embeddings

Cosine similarity



★ Semantic Fusion Strategies

Mean

Max

Softmax

Weighted

Frequency-aware semantic weighting
Entropy-aware uncertainty estimation

Redundancy-complementarity analysis (RQ3 · RQ4)



Region-based semantic similarity scoring

Semantic similarity computed over annotated object crops

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Experimental Evaluation



Semantic Recognition Metrics

AP_{seen} · AP_{unseen} · GZSD HM

Ablation Studies

Fusion strategy compare
Prompt count 1–10 effect
Prior-aware fusion impact

Robustness (RQ6)

Noisy/adversarial prompts
Perturbation stability

Semantic recognition scores

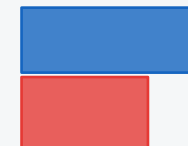
armadillo 0.94

mouser 0.87

chalice 0.71

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Results & Impact

↑ AP_{unseen}:
63.5 → 67.7 (+4.2 pts)Rare & unseen categories
vs. single-prompt
baseline

Ours (fusion)

Baseline

GZSD HM comparison

Fusion weight heatmap

Findings contribute to
Info Fusion + CV
communities